

OrbitDesk

Project Management System for NGOs

1. System Overview

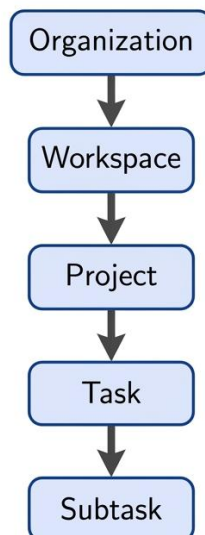
1.1 Introduction

OrbitDesk is a web-based project management platform purpose-built for nongovernmental organizations (NGOs). It unifies standard project workflows — Kanban boards, task hierarchies, and reporting — with NGO-specific financial governance: donor tracking, budget management, and grant compliance, so that program delivery and financial accountability live in one system.

This edition consolidates the role model into a single, consistent set of seven roles applied uniformly across every level of the hierarchy, and folds in eight items of client feedback covering leadership continuity, bulk member management, credential security, file handling, task-status accountability, team-replacement tracking, comment separation, and system-wide enum usage.

1.2 Hierarchical Structure

- **Organization** — the legal NGO entity or country office.
- **Workspace** — a department, programme area, or field office.
- **Project** — a funded initiative with budget and donor linkage.
- **Task** — a unit of deliverable work.
- **Subtask** — a component of a Task.

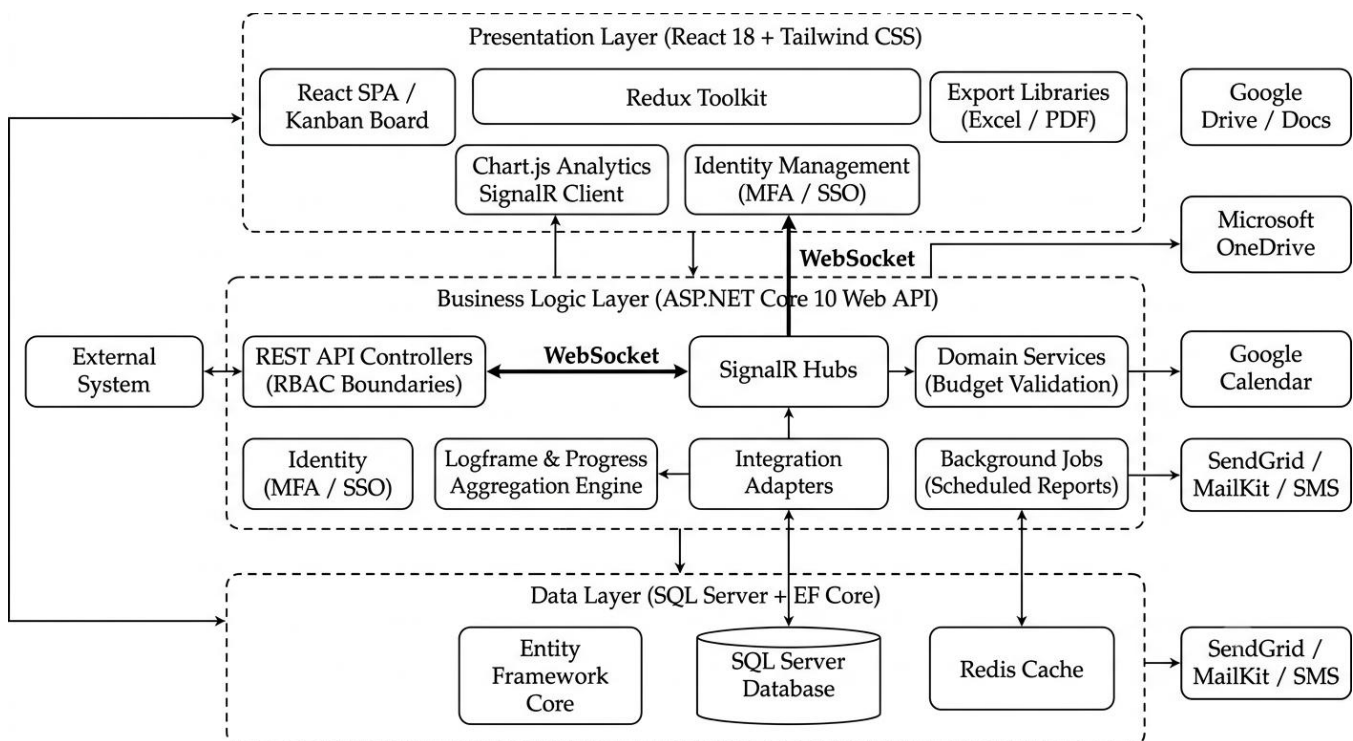


1.3 Core Philosophy

- **Real-world NGO applicability:** project workflows combined with donor tracking and financial reporting NGOs are contractually required to produce.
- **Accountability:** a complete, immutable audit trail — including who changed a Task's status and who replaced whom on a Project — and financial reporting suite.
- **Clarity of authority:** one consistent, enum-backed role model, not a different one per module.
- **Scalability:** a clean core architecture with NGO-specific modules layered on top.
- **Compliance by design:** donor reporting formats and grant conditions are first-class objects, not afterthoughts.

1.4 Key Differentiators

- **Unified RBAC:** seven roles — Owner, Admin, Coordinator, Manager, Finance Officer, Member, Viewer — applied consistently at Organization, Workspace, and Project scope, each backed by a database enum.
- **Team Management:** reusable Teams defined once per Workspace, assigned as a unit to any Project, with a full replacement/history trail.
- **Kanban Workflow:** hierarchical Tasks, Subtasks, and Microtasks with drag-and-drop and a per-change audit trail.
- **Real-time Collaboration:** live updates and in-app notifications via SignalR.
- **Financial Management:** donor tracking, multi-level budgets, expense approvals, grant compliance.
- **Volunteer Management:** distinct from paid staff, with hours logging for in-kind reporting.
- **Risk Register:** risk and issue management with Organization-level roll-up.
- **Gantt / Timeline:** task duration, dependencies, critical-path view, and postponement history for donor presentations.



2. Roles & Permissions (RBAC)

This is the single source of truth for authority in OrbitDesk. Every permission check in every module resolves against these seven roles — there are no module-specific role names elsewhere in the system.

2.1 Role

Role	Description & Authority
Owner	Ultimate authority over the Organization. Only role that can transfer ownership, permanently delete the Organization, or override any approval. Exactly one Owner per Organization at a time.
Admin	Manages Organizational setup: members, workspaces, roles, compliance profile, SSO/MFA policy, and consortium links. Cannot transfer ownership or delete the Organization.
Coordinator	Coordinates Projects and teams across a Workspace: assigns members to Projects, manages cross-project scheduling, owns the Risk & Issue Register, and assigns Volunteers.
Manager	Runs individual Projects day-to-day: creates/edits Projects and Tasks, manages the Kanban board, sets deadlines, and drafts budgets and expense sign-off within their assigned Project(s).
Finance Officer	Owns financial governance: budgets, expense approval, donor allocations, grant compliance, and financial dashboards. Read-only outside financial modules.
Member	Performs assigned work: updates status on Tasks/Subtasks/Microtasks assigned to them, comments, attaches files, logs hours. No management rights over others' work.
Viewer	Read-only access to whatever scope they are granted (Organization, Workspace, or Project). Cannot create, edit, approve, or delete anything.

2.2 Where Roles Apply

A role is always assigned at a specific scope (Organization, Workspace, or Project). A user can hold different roles at different scopes simultaneously

Hierarchy Level	How Roles Apply
Organization level	Owner and Admin are assigned here. A user's Organization role is their default (highest) permission floor.
Workspace level	Coordinator is typically assigned here to oversee all Projects within the Workspace; Viewer can also be scoped here for stakeholders who only need visibility into one department.
Project level	Manager, Finance Officer, Member, and Viewer are typically assigned here. A user may hold different roles on different Projects (e.g., Manager on Project A, Member on Project B).
Task level	Assignment only (Task Members) — not a distinct role. Task assignment inherits the assignee's Project-level role for permission checks.

2.3 Permission Matrix

"Full" = create, edit, delete, approve. "View*" = read-only, granted only if the Viewer has been explicitly scoped to that Organization/Workspace/Project. Parenthetical notes qualify the scope of the permission.

Module	Owner	Admin	Coordinator	Manager	Finance Officer	Member	Viewer
Organization Settings	Full	Full	View	View	View	None	View*
Member & Role Mgmt	Full	Full	Invite (Project)	Assign (Project)	None	None	None
Workspace Management	Full	Full	Edit	View	View	None	View*
Project Management	Full	Full	Full	Full (owned)	View	View (assigned)	View*
Team Management	Full	Full	Full	Assign (existing team)	View	View (own team)	View*
Task / Kanban	Full	Full	Full	Full (owned)	View	Edit (assigned)	View*
Risk & Issue Register	Full	View	Full	Edit (owned)	View	Create issue	View*
Budget Management	Full	View	View	Draft / Propose	Full	None	View*
Expense Tracking & Approval	Full (override)	View	None	Sign-off (2nd step)	Full (1st review)	Submit request	View*
Donor & Grant Management	Full	View	View	View	Full	None	View*
Reporting & Dashboards	Full	Full	View	View / Generate	Full (financial)	View (own)	View*
Volunteer Management	Full	Full	Full	Edit (assign)	View (in-kind)	View	View*
Comments (Project & Task)	Full	Full	Full (scope)	Full (owned)	View	Own (assigned)	View*
Audit Log	View	View	None	None	View (financial)	None	None
Security / SSO / MFA Policy	Full	Full	None	None	None	None	None

Rule of precedence: if a user holds conflicting roles at different scopes (e.g., Viewer at Organization level but Manager on one Project), the more specific (lower-level) scope wins for that Project; the Organization-level role remains the floor everywhere else.

2.4 Approval Workflow Ownership

- **Budget creation & revision:** drafted by Manager, approved by Finance Officer, visible to Owner/Admin/Coordinator.
- **Expense approval:** two-step — Finance Officer reviews first, Manager signs off second; Owner can override either step.
- **Grant compliance checklist:** owned end-to-end by Finance Officer; Coordinator and Manager can view and flag blockers.
- **Ownership transfer:** only the current Owner can initiate; requires confirmation and is written to the immutable Audit Log.

3. Functional Requirements

3.1 Organization Management

- Create Organization with name, description, logo, registration number, and country.
- Member invitation via email with expiring links and role pre-assignment.
- Ownership transfer with confirmation and Audit Log entry.
- Soft-delete with backup and restore capability.
- Multi-entity / consortium support: link partner Organizations for joint Projects.
- Organization-level compliance profile: registration documents, tax-exemption status, renewal dates with automated reminders.

Roles: Owner and Admin — Full. All other roles — View only, if scoped.

3.2 Workspace Management

- Full CRUD operations; archive Workspaces.
- Member management with Workspace-level role assignment.
- **Adding members uses a multi-select search picker**, not one-by-one entry: type-ahead search by name or email across the Organization's directory, tick any number of results, and add them all in a single action. This applies everywhere members are added to a Workspace, a Team, or a Project — essential once a Workspace needs many users added at once.
- Visibility settings: Public, Private, Restricted.
- Workspace dashboard: active Projects, tasks due, budget utilization.
- Optional Workspace-level budget ceiling capping all child Projects.

Roles: Owner and Admin — Full. Coordinator — Edit (day-to-day coordination). Manager/Finance Officer — View. Member — none. Viewer — View if scoped.

3.3 User Management & Authentication

- Secure registration and authentication; **Admin/Owner never sets or sees a password when creating a user.** User creation by Admin/Owner captures only name, email, and initial role — the system sends a tokenized, time-limited invite link, and the user sets their own password on first login. security requirement.
- Password reset is always self-service via a tokenized link, never performed on a user's behalf by an administrator.
- Profile management: name, photo, contact information, preferred language.
- Multi-level RBAC enforced at Organization, Workspace, and Project scope.
- Theme customization: light and dark modes.
- Multi-factor authentication (MFA) — required for Owner, Admin, and Finance Officer; recommended for all others.
- Single Sign-On (SSO) with Google Workspace.
- Session and device management: view and revoke active sessions.

Roles: Owner and Admin manage org-wide security policy (SSO/MFA enforcement) and initiate invitations, but never set credentials. All users manage their own profile, password, and sessions.

3.4 Project Management

- Project creation with title, description, start/end dates, budget, and donor linkage.
- **Project Lead assignment with a tracked tenure (start date → end date).** Because the lead changes over the life of a Project, each assignment is stored as its own dated record rather than a single fixed field: the current lead has an open-ended record (EndDate = NULL); reassigning the lead closes the old record with today's date and opens a new one, so the full leadership history is always queryable.
- Status lifecycle: Planning → Active → On Hold → Completed → Cancelled → Archived (enum).
- Team assignment with Project-scoped roles, including full mid-Project replacement tracking.
- **Timeline postponement is a first-class, auditable action**, not a silent edit: any change to a Project's end date — for example, following a team replacement — is recorded with the old date, the new date, the reason, and who requested/approved it
- Archive/delete with configurable retention period (soft-delete, recoverable).
- Project dashboard: financial overview, task completion rate, deadlines, risk flags, current lead, and postponement history.

3.4.1 Logframe / Results Framework

- Goal → Outcome → Output → Activity hierarchy per Project.
- Indicators per Output/Activity with baseline, target, actual value, and unit.
- Automatic progress aggregation from linked Tasks.
- Export in donor-preferred formats, linked to Grant Compliance reporting.

3.4.2 Risk & Issue Register

- Risk log: description, likelihood, impact score, mitigation plan, and owner.
- Issue log: problems already occurred, with resolution tracking and status.
- Roll-up view at Organization level for leadership review.

Roles: *Coordinator — Full across Projects in their Workspace. Manager — Full on owned Project(s), including initiating lead reassignment and postponement requests. Owner/Admin — Full everywhere, including approving postponements. Finance Officer/Viewer — View. Member — can create Issues, cannot edit Risk log.*

3.5 Team Management

A Team is a reusable, named group of Users defined once at Workspace level and then assigned as a unit to one or more Projects — instead of adding members to each Project one by one.

- Team creation within a Workspace: name, description, and a designated Team Lead.
- **Add or remove Team members via the same multi-select search picker described in** no one-by-one addition, even for large Teams.
- Assign an entire Team to a Project in a single action; each member's existing Project-scoped role (via RoleAssignment) still governs what they can do — Team assignment only bulk-adds people, it does not grant permissions on its own.
- **Mid-Project Team/member replacement is tracked explicitly.** If a Team or an individual assigned to a Project is swapped out before the Project finishes, the system closes the old assignment (RemovedAt) and links it to its replacement (ReplacedByTeamID), so it is always clear who was responsible for the work during any given period — and, any resulting date postponement is logged alongside it.
- Team roster view showing each member's current role, workload, and availability.
- Team workload dashboard: open/overdue Tasks aggregated across all members of the Team.
- Move or copy a Team from one Project to another when a program cycle ends or a new phase begins — recorded through the same replacement history.
- Archive a Team (soft-delete) while retaining its historical Project-assignment record for audit.

Roles: *Coordinator — Full (primary owner of Team formation, replacement, and cross-project coordination). Owner/Admin — Full. Manager — can assign an existing Team to their own Project and request a replacement, but cannot create/edit Teams outright. Finance Officer/Viewer — View. Member — View their own Team roster only.*

3.6 Task & Workflow Management

- Task creation with hierarchical Subtasks and Microtasks.
- Kanban board with drag-and-drop: To Do → In Progress → In Review → Blocked → Done (enum).
- **Every status change is attributed:** each transition logs who changed it, the old status, the new status, and the timestamp, visible in the Task's activity history. → client item 5.
- **Planned Deadline vs. actual CompletedDate are tracked separately.** When a Task is moved to Done, CompletedDate is set to the real date of completion. If CompletedDate falls before Deadline, the Task is flagged as finished ahead of schedule — the stored 'finish date' is always the true completion date, never the original deadline.
- Priority levels: Low, Medium, High, Urgent (enum).
- Due dates with automated reminders.
- Many-to-many Task assignment across team members.

- Comments **attachments with in-app open/preview**, and full activity history on every Task.
- **Attachments open inside OrbitDesk without requiring a download**: images, video, PDF, and Office documents (.docx, .xlsx, .pptx) render in an in-app viewer; download stays available as a secondary option. Files themselves are stored on disk/object storage — only the absolute file path and metadata (media type, MIME type, size) are persisted in the database.
- Cloud storage integration: Google Drive and OneDrive.
- Keyboard shortcuts for power users.
- Recurring Tasks (e.g., monthly distributions, quarterly check-ins).
- Task dependencies (finish-to-start, start-to-start) with critical-path highlighting.

3.6.1 Gantt / Timeline View

Projects can be visualized as a Gantt chart showing Task duration, dependencies, critical path, and any recorded postponements — a format frequently requested by donors and auditors.

Roles: Manager and Coordinator — Full on their scope. Member — Edit only Tasks assigned to them (status, comments, attachments); every edit they make to status is captured in TaskStatusHistory under their own UserID. Finance Officer/Viewer — View. Owner/Admin — Full.

3.7 Communication

- In-app notifications: assignments, mentions, due dates, status changes, lead reassignment, team replacement, and postponements.
- Real-time updates via SignalR.
- Email alerts: configurable as digest or immediate.
- In-app announcements at Organization or Workspace level.
- SMS alerts for critical events in low-connectivity areas.

Roles: All roles receive notifications relevant to their scope. Announcements can be posted by Owner, Admin, and Coordinator.

3.8 Reporting & Dashboards

- Task analytics: completion rate, on-time vs. late finish (Deadline vs. CompletedDate), cycle time, overdue counts.
- Financial analytics: budget vs. actual spend, donor contributions by Project.
- Chart types: bar, pie, burndown.
- Workload distribution view across team members.
- PDF and Excel export for all reports.
- Scheduled report delivery: auto-generated reports emailed on a recurring schedule.

Roles: Finance Officer owns financial reports; Owner/Admin see everything; Coordinator/Manager see operational reports for their scope; Member sees their own workload; Viewer sees whatever they are scoped to.

3.9 Financial Management (NGO-Specific)

3.9.1 Donor Management

- Donor profiles: contact details, type (institutional, foundation, individual, corporate — enum).
- Contributions: pledged vs. received amounts and payment schedules.
- Allocations: link contributions to specific Projects or grants.
- Donor communication log for relationship tracking.

3.9.2 Budget Management

- Budgets at Organization, Workspace, Project, and Task level with roll-up aggregation.
- Budget line items categorized by type: personnel, travel, supplies, overhead.
- Budget revisions with full version history and approval trail.
- Single-currency budgeting (multi-currency planned).

3.9.3 Expense Tracking

- Expenses linked to a Project and, optionally, to an individual Task.
- Receipt and invoice attachment required above a configurable threshold — uses the same in-app viewer as Task attachments.
- Multi-step approval workflow: Finance Officer review → Manager sign-off.
- Budget line validation: blocks or flags overspending in real time.

3.9.4 Financial Dashboards

- Income vs. expenses at Organization and Project level.
- Donor utilization reports summarizing spend per contribution.
- Burn-rate forecasting to project future spend.

3.9.5 Grant Management & Compliance Reporting

- Grant allocation and utilization tracking per Project.
- Grant conditions and compliance checklist: trackable items with due dates.
- Audit support package: one-click export of the full financial trail (budget versions, expenses, approvals, receipts, donor allocations) formatted for external audit.

Roles: Finance Officer — Full across all financial modules. Owner — Full with override authority.

Manager — drafts budgets and gives second-step expense sign-off within their Project.

Admin/Coordinator/Viewer — View. Member — none, except submitting an expense request where permitted.

3.10 Search

- Global search across Projects, Tasks, donors, and financial records, scoped by user permissions.
- Faceted filtering by type, date, status, and assignee.
- Saved searches for frequently used queries.

Roles: Search results are always filtered to what the requesting role is permitted to see — a Member never sees financial records in results, a Viewer never sees anything outside their granted scope.

3.11 Volunteer Management

- Volunteer registry: contact details, skills, availability, background-check status.
- Assign Volunteers to Tasks or events without issuing full user accounts (optional limited login).
- Volunteer hours logging for in-kind contribution reporting.

Roles: *Coordinator — Full (primary owner of volunteer assignment). Owner/Admin — Full. Manager — can assign Volunteers to their Project's Tasks. Finance Officer — View (for in-kind reporting). Member/Viewer — View.*

3.12 Comments (Project & Task, Kept Separate)

Comments are attached to exactly one of two entity types — a Project or a Task — never ambiguously to both. Each comment record carries an EntityType and EntityID, so Project-level discussion (overall scope, budget posture, donor relationship) is fully separate from Task-level discussion (how to complete a specific deliverable), while both are served by one reusable comment engine (threading via ParentCommentID, @mentions, edit history). → client item 7.

- Post, edit (within a time window), and soft-delete comments; edit history retained for audit.
- @mention support triggers a notification to the mentioned user.
- Project comments are visible to everyone scoped to that Project; Task comments are visible to everyone scoped to the parent Project, filtered by role as usual.

Roles: *Owner/Admin/Coordinator — Full within scope. Manager — Full on owned Project's comments (both Project- and Task-level). Member — post/edit own comments on Tasks assigned to them and on Projects they belong to. Finance Officer — View. Viewer — View*.*

4. Non-Functional Requirements

- **Performance:** fast page loads on broadband; quick API response times under normal load.
- **Security:** bcrypt password hashing; JWT-based auth; HTTPS enforced; RBAC checked at every API boundary against the seven-role model in §2. Encryption at rest for financial/donor records and in transit (TLS 1.2+). Field-level access restriction (e.g., a Manager sees budget totals, not vendor banking details). Passwords are always set by the account holder, never by an administrator (§3.3). Regular dependency and vulnerability scanning.
- **Usability:** responsive design for desktop and tablet. English and Amharic language support with a translation framework ready for additional languages. All fixed-choice inputs (role, status, priority, media type, etc.) render as enum-driven dropdowns/pickers, never free text — see §5.5.
- **Reliability:** structured error handling with no internal details exposed to end users. Automated daily backups with documented, tested restore procedures.
- **Scalability:** horizontal scaling at the application tier. Database designed for single-office to multi-country deployments, with proper indexing and read-replica support.
- **Data Governance:** configurable data retention policies (automatic archival, not deletion). Full data export capability for all records. Audit Log immutability — append-only, cannot be edited or deleted. TaskStatusHistory, ProjectLeadHistory, ProjectTeamHistory, and ProjectPostponements are likewise append-only.

5. Database Schema

5.1 Roles & Access (first-class)

Entity	Key Fields
Roles	RoleID, Name (enum: Owner / Admin / Coordinator / Manager / FinanceOfficer / Member / Viewer), Description — fixed, system-defined list.
RoleAssignments	AssignmentID, UserID, RoleID, ScopeType (enum: Organization / Workspace / Project), ScopeID — one row per role a user holds at a given scope; this is how one person can hold several roles at once.

5.2 Core Entities (updated)

Entity	Key Fields
Organizations	OrgID, Name, OwnerID, Budget, Currency, RegistrationNo. Base reporting currency stored here.
Users	UserID, Name, Email, PasswordHash (Nullable — stays NULL until first login), MFAEnabled. Roles resolved via RoleAssignments, never stored directly on the user.
UserInvitations	InvitationID, UserID, InvitedByUserID, Token, ExpiresAt, Status (enum: Pending / Accepted / Expired). Set by Admin/Owner at creation; only the invited user ever writes PasswordHash.
Workspaces	WorkspaceID, OrgID, Visibility (enum), BudgetCeiling. Optional budget cap.
Teams	TeamID, WorkspaceID, Name, Description, TeamLeadUserID, IsArchived. A reusable member grouping, defined once per Workspace and assigned as a unit to Projects.
Projects	ProjectID, WorkspaceID, Title, Status (enum), StartDate, EndDate, Budget, DonorID. Current Project Lead is derived from ProjectLeadHistory (row with EndDate IS NULL), not stored directly here.
ProjectLeadHistory	ProjectLeadHistoryID, ProjectID, UserID, StartDate, EndDate (NULL = current lead). Reassigning the lead closes the open row and inserts a new one.
Tasks	TaskID, ProjectID, Title, Status (enum), Priority (enum), Deadline (planned), CompletedDate (actual, Nullable), ParentTaskID. ParentTaskID enables hierarchy.
TaskStatusHistory	HistoryID, TaskID, OldStatus, NewStatus, ChangedByUserID, ChangedAt. Append-only.
Subtasks / Microtasks	SubtaskID, TaskID, Title, IsDone. Checklist items under Tasks.
Comments	CommentID, EntityType (enum: Project / Task), EntityID, UserID, Content, ParentCommentID (threading), CreatedAt, EditedAt. Replaces the old Task-only foreign key so Project and Task discussion never mix.
Attachments	AttachmentID, EntityType (Project / Task), EntityID, FileName, AbsoluteFilePath (physical path on disk/object storage), MediaType (enum: Image / Video / Document / Audio / Other), MimeType, FileSizeBytes, PreviewEnabled.
Notifications	NotificationID, UserID, Message, Channel (enum: in-app / email / SMS).
AuditLogs	LogID, Entity, Action, OldValues, NewValues, Timestamp, PerformedByUserID.

Entity	Key Fields
	Append-only, immutable.

5.3 Financial, Compliance & History Entities

Entity	Key Fields
Donors	DonorID, Name, Contribution, AllocatedProject, DonorType (enum: institutional / foundation / individual / corporate).
Expenses	ExpenseID, ProjectID/TaskID, Amount, Currency, Date, Description, ApprovalStatus (enum), ApprovedByFinanceOfficerID, SignedOffByManagerID.
BudgetRevisions	RevisionID, ProjectID, VersionNo, ApprovedByFinanceOfficerID, ApprovedDate.
GrantConditions	ConditionID, GrantID, Description, DueDate, Status.
Volunteers	VolunteerID, Name, Skills, Availability, BackgroundCheckStatus. Distinct from Users table.
RisksIssues	RiskID, ProjectID, Type (enum: Risk / Issue), Likelihood, Impact, Owner, Status.
Indicators	IndicatorID, ProjectID, OutputID, Baseline, Target, Actual, Unit. Logframe / M&E support.
ProjectTeamHistory	HistoryID, ProjectID, TeamID, AssignedAt, RemovedAt, ReplacedByTeamID. Records every Team/member swap on a Project before completion.
ProjectPostponements	PostponementID, ProjectID, OldEndDate, NewEndDate, Reason, RequestedByUserID, ApprovedByUserID, CreatedAt. Any timeline shift is logged here rather than silently overwriting Projects.EndDate.

5.4 Junction Tables

- RoleAssignments (UserID, RoleID, ScopeType, ScopeID) — replaces the old per-level *Members tables; also the mechanism for a person holding multiple roles.
- TeamMembers (TeamID, UserID, JoinedAt) — many-to-many membership in a Team, populated via the multi-select search picker.
- ProjectTeams (ProjectID, TeamID, AssignedAt) — current, active assignment of a Team to a Project; superseded rows move into ProjectTeamHistory on replacement.
- TaskMembers (TaskID, UserID) — many-to-many Task assignment.
- TaskDependencies (TaskID, DependsOnTaskID, DependencyType (enum)).
- ProjectDonors (ProjectID, DonorID, AllocatedAmount) — supports multiple donors per Project.

5.5 Enumerations (system-wide)

Every field in OrbitDesk that represents a fixed set of choices — not just Role — is implemented as a database-level enum, never free text. This keeps every dropdown, tag, and multi-role assignment constrained to valid values and makes RBAC and status logic queryable and reliable.

Enum	Values	Used In
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Enum	Values	Used In
RoleName	Owner, Admin, Coordinator, Manager, FinanceOfficer, Member, Viewer	Roles.Name
ScopeType	Organization, Workspace, Project	RoleAssignments.ScopeType
ProjectStatus	Planning, Active, OnHold, Completed, Cancelled, Archived	Projects.Status
TaskStatus	ToDo, InProgress, InReview, Blocked, Done	Tasks.Status, TaskStatusHistory
Priority	Low, Medium, High, Urgent	Tasks.Priority
MediaType	Image, Video, Document, Audio, Other	Attachments.MediaType
EntityType	Project, Task	Comments.EntityType, Attachments.EntityType
Visibility	Public, Private, Restricted	Workspaces.Visibility
DonorType	Institutional, Foundation, Individual, Corporate	Donors.DonorType
ApprovalStatus	Pending, Approved, Rejected	Expenses.ApprovalStatus
NotificationChannel	InApp, Email, SMS	Notifications.Channel
DependencyType	FinishToStart, StartToStart	TaskDependencies.DependencyType
RiskIssueType	Risk, Issue	RisksIssues.Type
InvitationStatus	Pending, Accepted, Expired	UserInvitations.Status

Any select/dropdown/tag field in the UI is generated from one of these enums, so a person with multiple roles, a Task moving between statuses, or a file being categorized by media type is always constrained to a valid, database-enforced value.

6. Technology Stack & Architecture

- **Frontend:** React 18, Tailwind CSS, Redux Toolkit, Chart.js.
- **Backend:** ASP.NET Core 10 Web API, C# 12, EF Core, SignalR.
- **Database:** SQL Server, with Redis cache for real-time and session data.
- **File storage:** attachments stored on disk/object storage; only the AbsoluteFilePath and metadata are persisted in SQL (§5.2).
- **Integrations:** Google Calendar, Google Drive, Microsoft OneDrive, SendGrid / MailKit.
- **Reporting:** Excel and PDF export libraries.

Architecture is layered: Presentation (React SPA, SignalR client, Identity/MFA/SSO UI, in-app file viewer) → Business Logic (REST API controllers enforcing RBAC boundaries, SignalR hubs, domain services for budget validation and status-change auditing, background jobs for scheduled reports) → Data (EF Core over SQL Server,

Redis cache). Every API controller checks the caller's RoleAssignment for the relevant scope before executing a request, and every state-changing action against a Task, Project Lead, Team assignment, or Project timeline writes its corresponding history row in the same transaction.

7. Testing Strategy

7.1 Test Levels

- **Unit:** business logic — budget validation, approval state transitions, currency conversion utilities, Task-status transition rules, lead/team history closure logic.
- **Integration:** API endpoints against a test database; RBAC boundary enforcement for all seven roles at all three scopes; multi-select bulk-add endpoints; invitation-token flow.
- **End-to-End (E2E):** critical journeys such as Create Project → Assign Budget → Record Expense → Generate Donor Report, and Replace Team → Postpone Deadline → Notify Stakeholders, exercised once per relevant role (Manager drafts, Finance Officer approves, Owner overrides).
- **UAT:** actual NGO staff — Coordinators, Managers, Finance Officers — validate usability and field relevance, including the new multi-select add-member flow and in-app file preview.

7.2 Specialized Testing

- Load tests to verify performance targets (50+ concurrent users).
- Security tests: authentication bypass attempts, RBAC boundary checks per role, verification that no code path allows an Admin/Owner to set another user's password, dependency vulnerability scans.
- Backup and restore drills with verified recovery time.

8. Future Enhancements

Excluded from current scope, planned for subsequent releases:

- **Multi-Currency Accounting:** budget in donor currency, expenses in local currency, exchange-rate-on-date conversion, configurable monthly reference rate.
- **Beneficiary Data & Monitoring:** aggregate beneficiary counts per activity/output, with optional API hooks to dedicated M&E or case-management systems.
- **Slack / Teams Integration:** push notifications and alerts directly into team communication platforms.
- **Custom Roles:** org-level ability to define additional fine-grained roles beyond the base seven, for organizations with unusual governance structures.